ICR-Net: Robust Deepfake Detection under Temporal Corruption

Chan Park, Hyeongjun Choi, Muhammad Shahid Muneer, Binh Minh Le, and Simon S. Woo*

College of Computing and Informatics, Sungkyunkwan University, Suwon, South Korea
{pchan1018, junhjun, shahidmuneer, bmle, swoo}@g.skku.edu

Abstract. Deepfake video detection aims to distinguish AI-generated facial forgeries from authentic videos. While recent methods show strong performance under spatial corruptions, their robustness against temporal corruptions remains largely unexplored. In practical streaming scenarios, network disruptions such as packet loss, bit errors, and aggressive compression induce temporal degradations that current benchmarks do not cover. To bridge this gap, we introduce the DeepFake Temporal Corruption Benchmark (DF-TCB), built on FaceForensics++ and DFDC datasets with diverse corruption types and severities. Our analysis reveals that existing detectors are highly fragile under these disruptions. We therefore propose ICR-Net, which predicts frame reliability and selectively corrects corrupted features. By leveraging clean-corrupted contrastive learning, it extracts corruption-invariant, class-separable representations. We demonstrate that ICR-Net achieves state-of-the-art robustness and cross-dataset generalization under diverse temporal corruptions.

Keywords: Deepfake Video · Temporal Corruption · Model Robustness

1 Introduction

Deepfake video detection is an important task for ensuring digital media integrity. Consequently, developing robust deepfake detection methods [38] has become critical research. While existing methods have achieved remarkable performance advancements, their robustness has been evaluated exclusively against spatial corruptions [9,43,32]. This leaves their vulnerability to temporal corruptions largely unexplored. Such an oversight is particularly critical in live-streaming [26], where packet loss, bit errors, and compression intermittently degrade the stream, causing abrupt frame transitions and destroying spatial forgery cues.

Existing detectors fall into frame-based [18,3,34] and video-based [32,8] methods. Frame-based approaches operate on individual frames, primarily exploiting spatial artifacts such as blending boundaries and texture anomalies. However, temporal corruptions can easily erase these cues [15], rendering simple frame-wise score aggregation highly unreliable. Conversely, video-based methods multi-frame

* Corresponding author

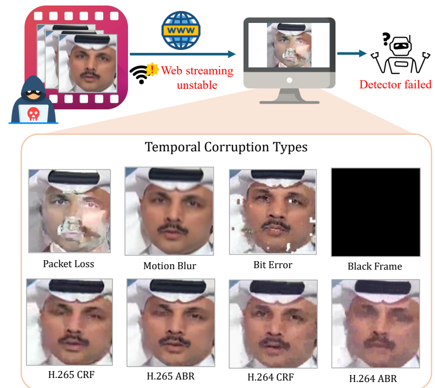

Fig. 1. Deepfake videos with temporal corruptions in the real-world scenario.

clips to encode spatio-temporal dynamics. Recent 3D-CNN detectors [43,32] implicitly assume smooth motion and uniformly reliable frames across the clip. When temporal corruptions violate these assumptions, these models struggle to disentangle malicious manipulations from benign network corruptions, often producing false positives on authentic videos. Ultimately, current detectors lack the explicit mechanisms to estimate frame reliability or correct degraded frames. This reveals a fundamental gap in temporal robustness, prompting three critical research questions: **RQ1.** How susceptible are current deepfake detectors to temporal corruptions? **RQ2.** How do these corruptions impact cross-dataset generalization? **RQ3.** How can we architect a framework that achieves temporal resilience without compromising clean-video accuracy?

To answer these questions, we introduce the DeepFake Temporal Corruption Benchmark (DF-TCB). Built upon the widely adopted FaceForensics++ (FF++) [28] and DFDC [5], DF-TCB evaluates temporal robustness across eight realistic transmission and compression artifacts at three distinct severity levels, yielding the corrupted splits FF++-C and DFDC-C. Furthermore, we propose ICR-Net, a novel framework designed to defend against these temporal degradations. ICR-Net ensures reliability-aware detection by predicting a per-frame integrity score and selectively applying residual corrections to low-integrity frames, thus preserving spatial forensic cues while suppressing temporal noise. Finally, utilizing paired clean and pre-generated corrupted clips, ICR-Net employs contrastive learning to extract corruption-invariant, class-separable representations. This alignment effectively reduces the feature-space confusion between benign corruption and actual forgery traces, substantially improving cross-dataset robustness across diverse corruption types, all while preserving peak performance on clean videos. Our main contributions are summarized as follows[1]:

- We introduce DF-TCB, the first benchmark dedicated to evaluating temporal robustness in deepfake detection.
- We propose ICR-Net, a reliability-aware framework that integrates temporal integrity assessment, frame-selective correction, and clean-corrupted contrastive learning.
- We demonstrate that ICR-Net achieves state-of-the-art robustness and cross-dataset generalization under temporal corruptions, without compromising accuracy on clean videos.

[1] Our code is available at https://github.com/Ckck12/ICR-Net.

2 Related Work

2.1 Frame-Based Deepfake Detection

Early deepfake detectors treat the task as frame-level classification and mainly exploit spatial artifacts such as local inconsistencies, global textures, and multi-scale anomalies in facial regions [7,25,36,20,31]. Subsequent works further leverage frequency-domain cues introduced by up-sampling and combine spectral and RGB features to capture complementary traces [6,27,16,19,21]. Data augmentation has also been used to improve generalization, for example via attention-guided perturbations, boundary-aware supervision, and self-blended or pseudo-deepfake faces [41,17,42,29]. However, by operating exclusively on individual frames, these methods are inherently limited in video settings.

2.2 Video-Based Deepfake Detection

Video-based methods extend frame-level detectors by modeling temporal dynamics over clips. Early works adapt architectures or training strategies to jointly learn spatial and temporal information, for example by balancing spatial-temporal weights, reshaping clips into thumbnail layouts, or fusing domains via joint 2D CNNs and dual-branch designs [32,33,8]. Other approaches explicitly target temporal consistency or region-/modality-specific cues, such as temporal convolutional networks with minimal spatial reliance, lip-motion-based detectors, and audio-visual self-supervised models [43,9,10]. However, these detectors do not explicitly address robustness to temporal corruptions.

2.3 Robustness of Deepfake Detection

FaceForensics++ [28] and DeeperForensics-1.0 [13] provide benchmark datasets with compression and synthetic distortions, enabling systematic evaluation under spatial corruptions. Existing works assess robustness to frame-level degradations such as saturation and contrast changes, Gaussian noise and blur, pixelation, JPEG compression, downscaling, and pixel dropout [35,40]. Consequently, these evaluation protocols have become standard in deepfake detection. Robustness evaluations of video-based detectors, however, still primarily focus on frame-wise corruptions [9,32,43]. In practice, temporal corruptions arise dynamically during encoding and streaming. While studies in video classification and action recognition [37,39] show that such disruptions can severely degrade performance, their critical impact on deepfake detection remains an open challenge.

3 DF-TCB Benchmark Creation

3.1 Benchmark Setup

We model temporal corruptions as localized degradations occurring within fixed-length video clips. Let $V = \{I_t\}_{t=1}^{T}$ represent an original clip consisting of T

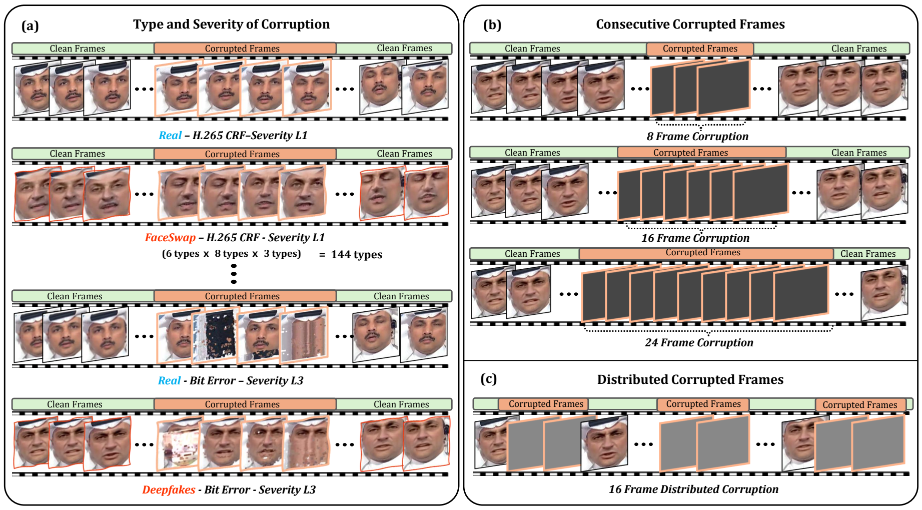

Fig. 2. Overview of the proposed DF-TCB. Based on the real and five manipulation types in FF++, we introduce (a) eight temporal corruption types, and evaluate scenarios with (b) consecutive and (c) distributed corrupted frames.

frames, with a ground-truth label $y \in \{0, 1\}$ denoting real and fake videos, respectively. Given a corruption operator $c(\cdot; l)$ with severity level l, and a subset of target frame indices $m \subset \{1, \ldots, T\}$, the resulting corrupted clip $\tilde{V} = \{\tilde{I}_t\}_{t=1}^{T}$ is defined as:

$$\tilde{I}_t = \begin{cases} c(I_t; l), & t \in m, \\ I_t, & \text{otherwise.} \end{cases} \tag{1}$$

Unless otherwise specified, we use 32-frame clips sampled with a fixed stride of 8. We focus on partial temporal corruptions as the main scenario. For the main setting, m forms a *consecutive* window with $|m| \in \{8, 16, 24\}$ within each clip.

3.2 Temporal Corruptions in Videos

We consider eight realistic temporal corruptions that can arise along the video pipeline, from acquisition through processing to transmission. We denote the corruption set by $c = \{\text{Black Frame}, \text{Motion Blur}, \text{Packet Loss}, \text{Bit Error}, \text{H.264 CRF}, \text{H.264 ABR}, \text{H.265 CRF}, \text{H.265 ABR}\}$ and use three severity levels $l \in \{1, 2, 3\}$:

- **Black Frame** [2]: Transient signal loss replacing selected frames with all-black pixels (levels: 8, 16, and 24 frames).
- **Motion Blur** [12]: Fast motion artifacts, temporally averaged over a sliding window (sizes: 3, 7, and 13).
- **Packet Loss** [37]: Network drops simulated via a virtual link (drop rates: 1%, 3%, and 5%).
- **Bit Error** [14]: Bitstream corruption causing blocking and smearing, simulated by bit flips (frequencies: 1 per 100,000, 30,000, and 10,000).

- **H.264 CRF** [22]: H.264 compression using Constant Rate Factor via libx264[2] (CRF values: 23, 37, and 51).
- **H.264 ABR** [30]: H.264 compression using Average Bitrate (re-encoded at 1/2, 1/8, and 1/32 of original bitrate).
- **H.265 CRF** [1]: The HEVC counterpart to H.264 CRF (CRF values: 27, 39, and 51).
- **H.265 ABR** [23]: The HEVC counterpart to H.264 ABR (re-encoded at 1/2, 1/8, and 1/32 of original bitrate).

3.3 DeepFake Temporal Corruption Benchmark

We construct DF-TCB using two standard deepfake datasets: FF++ [28] and DFDC [5]. Following the official FF++ protocol, we apply the aforementioned corruptions to create FF++-C for intra-dataset evaluation. Similarly, we corrupt the DFDC test set to generate DFDC-C, for cross-dataset robustness evaluation. By default, we apply 16-frame consecutive corruption within each 32-frame clip across all corruption types. Specifically, Packet Loss is simulated over a controlled Mininet topology and UDP transport[3], while codec-based corruptions (H.264/H.265 CRF and ABR), Bit Error, and Motion Blur are produced using FFmpeg[4]. We also introduce *distributed* variants for Black Frame and Motion Blur to probe sensitivity to corruption patterns (Sec. 5.2). In these variants, corrupted frames are scattered rather than contiguous to provide a more challenging testbed.

3.4 Preliminary Robustness Evaluation

Evaluation Protocol. We first evaluate the temporal robustness of nine representative deepfake detectors on DF-TCB. These include six frame-based methods (denoted with #): FFD [4], F3-Net [27], SPSL [19], SRM [11], CORE [24], and Effort [34], and three video-based methods (denoted with $*$): FTCN [43], STIL [8], and AltFreezing [32]. All models are trained on clean FF++ and evaluated in two scenarios: (i) intra-dataset, using FF++ and FF++-C, and (ii) cross-dataset, using DFDC and DFDC-C. We report video-level accuracy (ACC) aggregated by corruption type and severity.

Results. Fig. 3 presents the robustness of existing detectors on DF-TCB. In the intra-dataset setting (top), all detectors achieve near-perfect accuracy on clean FF++, but their performance degrades sharply under temporal corruptions. Among frame-based methods, Effort [34] exhibits the most significant average decline of 22.48%. Video-based methods show even greater vulnerability. For instance, FTCN [43] plummets by 38.72% from its 99.65% baseline on clean data. These results show that both frame- and video-based detectors are highly sensitive

[2] https://www.videolan.org/developers/x264.html
[3] https://mininet.org/
[4] https://ffmpeg.org/

to temporal disruptions, regardless of their performance on clean samples. The impact is even more severe in the cross-dataset setting (bottom). While models trained on clean FF++ maintain 64.76–84.15% accuracy on clean DFDC, even mild temporal corruptions reduce all detectors to nearly 50% accuracy on DFDC-C. This suggests that temporal corruptions not only degrade in-distribution performance but also effectively destroy cross-dataset generalization, highlighting the urgent need for temporally robust deepfake detectors.

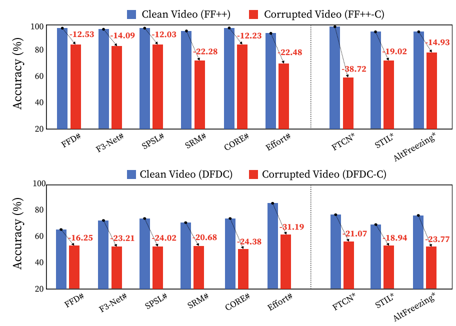

Fig. 3. Robustness of existing detectors against temporal corruptions.

4 Method

4.1 Problem Setup

To explicitly disentangle temporal corruption from malicious manipulation, ICR-Net leverages a paired learning strategy using clean and corrupted versions of the same video. Let $V \in \mathbb{R}^{T\times H\times W\times 3}$ be a video clip with a ground-truth label $y \in \{0, 1\}$. For each V, we construct a clean and a corrupted counterpart, denoted as $(V_{clean}, V_{corrupt})$. Specifically, we define these pairs for real and fake classes as $(V_{R\_clean}, V_{R\_corrupt})$ and $(V_{F\_clean}, V_{F\_corrupt})$, respectively. As illustrated in Fig. 4 (Step 1), all clips are first processed by a shared spatial encoder E_S, yielding a sequence of frame embeddings $S = [s_1, \dots, s_T] \in \mathbb{R}^{T\times D}$. These embeddings serve as the foundation for our integrity assessment and feature correction modules.

4.2 Temporal Integrity and Correction

The Temporal Integrity and Correction module (Step 2) restores corrupted features by identifying unreliable frames and estimating their clean counterparts. To achieve this, we employ a dual-branch architecture: a GRU-based predictor for global integrity assessment and a 1D-CNN branch for local residual estimation.

Temporal Prediction and Integrity Assessment. We first model the expected temporal regularity of the video to quantify frame reliability. We employ a GRU to predict the t-th embedding based on the preceding context:

$$h_t = \mathrm{GRU}(s_t, h_{t-1}; \theta_W), \quad \hat{s}_t = g(h_{t-1}). \tag{2}$$

The prediction error $e_t = s_t - \hat{s}_t$ captures the deviation from normal temporal dynamics. This error is mapped to an integrity score $\alpha_t \in (0, 1]$ using an exponential kernel:

$$\alpha_t = \exp(-\lambda_\alpha \|e_t\|_2). \tag{3}$$

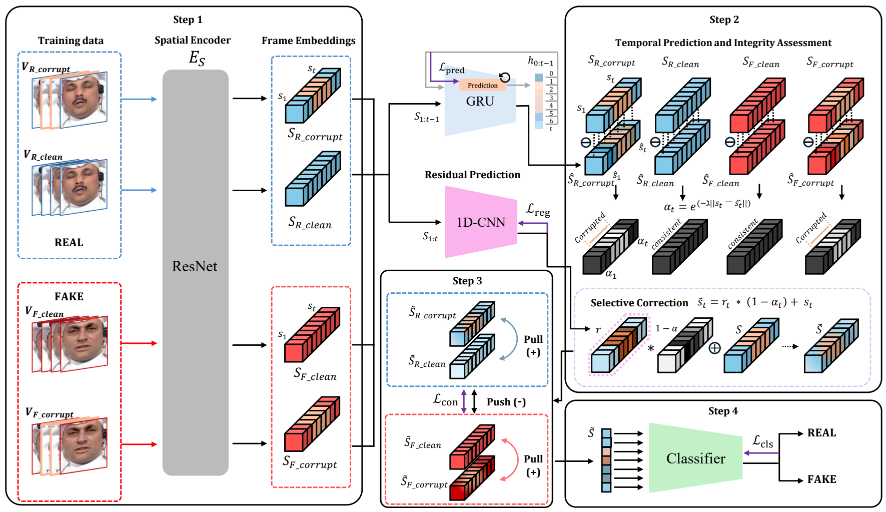

Fig. 4. Overview of our ICR-Net framework for temporal robustness.

Under this formulation, temporally consistent frames (small $\|e_t\|$) yield $\alpha_t \approx 1$, while corrupted frames with high irregularity (large $\|e_t\|$) result in lower scores. To ensure the GRU effectively captures legitimate video dynamics, it is optimized via a self-supervised prediction loss:

$$\mathcal{L}_{\text{pred}} = \frac{1}{T-1} \sum_{t=2}^{T} \|e_t\|_2^2. \tag{4}$$

Residual Prediction and Selective Correction. While the integrity score identifies where corruption occurs, a parallel 1D-CNN branch determines how to rectify it. By capturing local temporal patterns, the 1D-CNN estimates a residual r_t for each frame. This residual represents the necessary shift to project a corrupted embedding back toward the clean manifold. To prevent unnecessary alternations to reliable frames, we apply an integrity-weighted regularization loss:

$$\mathcal{L}_{\text{reg}} = \frac{1}{T} \sum_{t=1}^{T} \alpha_t \|r_t\|_2. \tag{5}$$

The original embedding s_t and the predicted residual r_t are then fused via an integrity-gated mechanism:

$$\tilde{s}_t = (1-\alpha_t) r_t + s_t. \tag{6}$$

This formulation ensures that corrections are primarily applied to low-integrity frames, while high-integrity frames are preserved, maintaining their original forensic cues.

4.3 Contrastive Representation Alignment

To ensure the model learns corruption-invariant yet class-separable features, we apply a contrastive learning strategy (Step 3). Given a batch of N quartets, each consisting of $(V_{R\_clean}, V_{R\_corrupt}, V_{F\_clean}, V_{F\_corrupt})$ derived from the same source, let z_i be the pooled clip-level embedding. We treat the clean and corrupted counterparts of the same video as a positives pair (z_i, z_i^+). For negatives, we utilize the two views (clean and corrupted) of the opposite label within the same quartet, denoted as $\mathcal{N}(i)$. We minimize an InfoNCE-style loss:

$$\mathcal{L}_{\text{con}} = -\frac{1}{4N} \sum_{i=1}^{4N} \log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\exp(\text{sim}(z_i, z_i^+)/\tau) + \sum_{z_n \in \mathcal{N}(i)} \exp(\text{sim}(z_i, z_n)/\tau)}, \quad (7)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is a temperature parameter. This objective aligns clean and corrupted representations of the same class, forcing the model to ignore transient temporal artifacts while preserving forgery traces.

4.4 Joint Optimization

The refined clip-level features are finally passed to a linear classifier to predict the forgery probability $\hat{y}_i = \sigma(W_c \tilde{s}_i + b_c)$, where $\tilde{s}_i$ is the pooled feature of the i-th clip (Step 4). We employ the standard binary cross-entropy loss to supervise the forgery detection task:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^{N} \Big[y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \Big], \quad (8)$$

The final training objective $\mathcal{L}$ is formulated as a multi-task loss that integrates the specific goals of each module:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{pred}} + \lambda_2 \mathcal{L}_{\text{reg}} + \lambda_3 \mathcal{L}_{\text{con}} + \lambda_4 \mathcal{L}_{\text{cls}}, \quad (9)$$

where we set $\lambda_1 = 1.0$, $\lambda_2 = 0.01$, $\lambda_3 = 0.5$, and $\lambda_4 = 1.0$ across all experiments. This weighting scheme ensures that the primary classification and self-supervised prediction tasks drive the optimization, while the regularization term remains at a smaller scale to avoid over-correcting reliable frames.

5 Experiment

5.1 Experimental Results

Evaluation Protocol. We investigate whether deepfake detectors can achieve enhanced robustness through explicit training on corrupted data. Following the protocol in [37], we adopt a corruption-aware training setting. For each training video, the dataset includes both a clean clip and a corresponding clip affected by a single temporal corruption at severity level 3. This protocol is applied uniformly to all baseline models and ICR-Net to ensure a fair comparison. We evaluate temporal robustness across eight corruption types in intra- and cross-dataset settings. We report performance for both clean and corrupted videos, with the latter representing the mean accuracy across three severity levels.

Table 1. Intra-dataset temporal corruption robustness of deepfake detectors.

	Model	Clean Frame	Black Frame	Motion Blur	Packet Loss	Bit Error	H.264 CRF	H.264 ABR	H.265 CRF	H.265 ABR
Frame-based	FFD [4]	98.14	85.89	91.23	90.98	90.83	91.80	88.70	91.44	90.63
	F3-Net [27]	98.10	85.84	88.16	90.22	90.47	90.47	88.33	89.21	92.05
	SPSL [19]	**98.29**	84.76	91.24	86.94	91.84	85.64	85.31	91.89	91.81
	SRM [11]	97.87	86.11	88.27	90.85	92.09	91.45	91.17	90.04	90.21
	CORE [24]	98.27	85.23	85.14	90.18	91.43	86.22	87.93	89.10	90.66
	Effort [34]	94.40	84.36	85.52	86.75	87.26	85.80	85.61	85.52	86.67
Video-based	FTCN [43]	87.62	86.37	82.64	82.97	82.65	83.48	83.69	83.69	83.54
	STIL [8]	97.35	86.00	91.31	86.83	89.58	88.35	89.15	92.16	89.28
	AltFreezing [32]	97.21	91.54	90.71	85.73	90.77	89.40	**94.58**	92.55	90.47
	ICR-Net *(ours)*	97.86	**94.92**	**94.88**	**96.03**	**96.67**	**96.55**	92.93	**97.50**	**95.83**

Table 2. Cross-dataset temporal corruption robustness of deepfake detectors.

	Model	Clean Frame	Black Frame	Motion Blur	Packet Loss	Bit Error	H.264 CRF	H.264 ABR	H.265 CRF	H.265 ABR
Frame-based	FFD [4]	61.41	50.32	50.12	50.49	50.22	50.38	50.19	51.02	49.73
	F3-Net [27]	72.02	49.93	49.77	50.01	49.92	50.13	49.68	49.33	49.59
	SPSL [19]	72.15	50.41	50.18	50.25	49.97	50.03	49.83	50.31	50.28
	SRM [11]	69.91	50.11	50.34	49.95	50.21	49.87	49.72	49.54	49.61
	CORE [24]	71.28	50.23	49.91	50.17	50.33	49.88	49.52	49.78	49.44
	Effort [34]	**79.85**	55.13	54.88	54.02	54.71	54.96	53.87	55.43	53.69
Video-based	FTCN [43]	61.21	53.11	52.89	53.39	54.12	53.02	52.07	53.34	53.55
	STIL [8]	67.71	49.15	49.98	48.50	50.05	48.61	48.01	50.24	49.82
	AltFreezing [32]	72.95	51.82	50.73	51.23	50.15	51.35	49.22	57.11	51.47
	ICR-Net *(ours)*	78.56	**58.14**	**59.23**	**59.51**	**58.98**	**59.12**	**57.98**	**59.88**	**58.33**

Intra-Dataset Results. Table 1 reports the results on FF++ and FF++-C. While all baseline detectors benefit from corruption-aware training and show improved robustness on FF++-C, this gain often comes at the cost of degraded accuracy on clean FF++. Despite this augmentation, a significant accuracy gap persists between clean and corrupted inputs, indicating that standard architectures lack explicit mechanisms to address temporal degradations. In contrast, ICR-Net achieves superior robustness across most corruption types while maintaining high accuracy on clean videos, demonstrating its ability to effectively disentangle transient noise from intrinsic forensic cues.

Cross-Dataset Results. The true challenge of temporal robustness lies in cross-dataset generalization to unseen environments. As shown in Table 2, most existing detectors suffer substantial degradation on DFDC-C despite corruption-aware training. Their robustness remains significantly lower than the in-distribution results observed in Sec. 3.4. ICR-Net consistently outperforms all baselines across all eight corruption types by a substantial margin. This consistent superiority suggests that our model learns generalized, corruption-invariant representations rather than merely memorizing training-specific artifacts.

Table 3. Ablation study of loss components in ICR-Net.

$\mathcal{L}_{\text{pred}}$	$\mathcal{L}_{\text{reg}}$	$\mathcal{L}_{\text{con}}$	Clean	Corrupted
×	×	×	91.57	86.38
✓	×	×	96.50	88.38
×	✓	×	82.54	77.61
×	×	✓	95.31	85.20
✓	✓	×	97.33	89.96
×	✓	✓	84.33	78.14
✓	×	✓	96.55	91.15
✓	✓	✓	**97.86**	**95.67**

Table 4. Comparison under various corruption patterns on FF++-C.

	Model	Black Frame (#) 8	16	24	Motion Blur (#) 8	16	24
Frame-based	FFD [4]	97.62	83.33	83.33	98.49	91.34	90.33
	F3-Net [27]	97.38	83.33	83.33	97.21	91.84	90.25
	SPSL [19]	**98.93**	83.33	83.33	98.33	91.17	**91.72**
	SRM [11]	96.07	92.74	16.67	97.06	88.78	87.35
	CORE [24]	97.30	96.67	83.33	98.58	85.16	88.15
	Effort [34]	83.26	84.76	83.33	95.36	83.93	86.90
Video-based	FTCN [43]	85.86	83.33	83.33	**99.65**	78.33	81.23
	STIL [8]	97.62	62.50	23.93	95.71	92.27	91.28
	AltFreezing [32]	96.43	91.51	88.87	96.90	94.88	90.60
	ICR-Net (*ours*)	98.21	**97.12**	**91.53**	97.61	**95.48**	91.17

5.2 Further Analysis

Ablation of Training Objectives. Table. 3 evaluates the contribution of each loss term. While classification loss $\mathcal{L}_{\text{cls}}$ (Row 1) defines the baseline, adding prediction loss $\mathcal{L}_{\text{pred}}$ (Row 2) only enhances clean performance. Using regularization loss $\mathcal{L}_{\text{reg}}$ alone (Row 3) is detrimental as the model trivially minimizes integrity scores, and contrastive loss $\mathcal{L}_{\text{con}}$ (Row 4) provides only modest gains without reliability modeling. The integration of these components yields significant improvements. Combining $\mathcal{L}_{\text{pred}}$ with $\mathcal{L}_{\text{reg}}$ (Row 5) stabilizes feature correction, while adding $\mathcal{L}_{\text{con}}$ (Row 7) promotes corruption-invariant representations. Ultimately, the full model (Row 8) achieves peak accuracy on both clean (97.86%) and corrupted (95.67%) data, proving that temporal prediction, selective correction, and contrastive alignment are all indispensable for robust detection.

Resilience to Corruption Dynamics. We further investigate how corruption patterns and magnitudes affect detection robustness. In the distributed setting, corruptions are applied in non-overlapping 4-frame blocks randomly scattered within each clip, totaling 8, 16, or 24 corrupted frames. As shown in Table. 4, existing methods suffer large performance drops as the number of corrupted frames increases, whereas ICR-Net demonstrates superior resilience, maintaining high accuracy with much slower degradation. This indicates that ICR-Net effectively handles varying patterns and intensities of scattered temporal corruption.

6 Conclusion

In this work, we introduce DF-TCB, the first benchmark to evaluate deepfake detector vulnerability to real-world temporal corruptions. Our evaluation reveals a critical gap in current forensics, as existing detectors suffer significant performance collapses under temporal degradations. We address this through ICR-Net, a framework that substantially enhances robustness and cross-dataset generalization while preserving accuracy on clean videos. By bridging the gap between laboratory settings and noisy streaming environments, this work establishes a necessary foundation for developing more resilient deepfake detection systems.

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